

Democracy and Bilateral Trade Flows*

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October 3, 2024

Preliminary and incomplete, please do not cite or circulate

Abstract: We study the effect of democratic regime change on bilateral trade flows in the post-WWII period, extending the standard structural gravity model empirics to 'heterogeneous gravity' estimated at the country-pair level. Our implementation mimics recent contributions to the heterogeneous difference-in-differences literature by augmenting the estimation equation with common factor proxies extracted from the sample of never-democratisers. This setup flexibly accounts for multilateral resistance and selection into democratic regime change while enabling us to study the effect of country-specific (monadic) variables. We find strong and lasting average effects of around 40% higher exports after two decades in democracy. The magnitude of the effect varies by regime status of the destination country, with exports to democracies 30 percentage points higher than those to autocracies. While democratic regime change had little lasting effect during the Cold War period, the results for data after 1989 suggest more modest long-run effects of around 25% after two decades. We then dig deeper into the heterogeneity of the democratic dividend for exports by estimating treatment effects for samples of treated and control countries split by favourable and unfavourable geography, legal origin, culture, or history. We show that in the 1950-2014 sample, legal origin aside, democratic regime change leads to unequal benefits in the form of higher exports for countries with favourable 'deep determinants'. In a separate analysis we find that the unequal benefits of regime change are more marked when we exclude the ideology-driven Cold War era.

Keywords: trade gravity model, heterogeneity, democratic regime change, panel data, interactive fixed effects, deep determinants of comparative development

JEL codes: F13, F14, P16, C23

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1 Introduction

The importance of trade for economic development is widely recognised among economists and policymakers alike — trade flows are associated with efficiency gains, technology transfer, and increased innovation activities. Traditionally, despite a lack of theoretical foundations, the applied literature studying the flow of goods (and, by extension, of investments, equity, migration, or remittances) adopted a ‘gravity model’ approach, which links flows to distance and economic mass (GDP) as well as trade ‘frictions’ (e.g. free trade agreements). Recent decades have seen two new developments in this literature: first, the gravity model has been considerably strengthened by theoretical work deriving it from a wide class of microeconomic principles, and by contributions to empirical methodology which have prescribed what the gravity specification should look like (in terms of variables and fixed effects to be included), and how it should be estimated to address a range of econometric challenges. Modern gravity empirics recognise that trade flows between countries cannot be studied in isolation but must account for the structure of the global trade network and the position of the trade partners within it — the force field of ‘multilateral resistances’ (MRTs). The second development is a direct consequence of this methodological progress: to adequately address the challenges of accounting for MRTs the ‘gold standard’ empirical gravity model rules out the investigation of country-specific (monadic) variables, i.e. determinants that vary at the country level such as institutions, exchange rate arrangements, or macroeconomic policy. Instead, modern gravity analysis is focused on a narrow set of variables defined at the country-pair (dyadic) level, such as regional trade agreements or currency unions.

Therefore, the empirical path taken by the gravity literature to achieve credibility and rigour is also its demonstrable weakness, an empirical straitjacket of sorts: identification is achieved by focusing on the narrow set of dyadic policy instruments, which represent a minority of the traditional levers that a country may use to become more integrated into the world economy. A second manifestation of the empirical straitjacket relates to the economic interpretation of the results derived from a structural gravity regression model: Is the estimated effect from the best practice model specification *common* to all countries in the sample, or does it represent an *unweighted or weighted average* (and if so, what are the weights)? These questions have obvious policy relevance, in that the global average effect is arguably at best meaningless and at worst seriously misleading for policy decisions in an individual country if this average hides substantial differences in the effect across countries.

In this paper, we propose an empirical approach to break this straitjacket: building on recent contributions to macro panel econometrics we develop a methodology to investigate the gravity relationship at the country-pair level, account for MRTs, and estimate the model in its multiplicative form, while enabling us to investigate the effect of monadic variables on trade flows. We adopt a multifactor error structure to capture the MRTs, treating the factors as nuisance terms in the

same fashion as the standard pooled gravity model employs a myriad of fixed effects. There are two alternative ways to proxy the unobserved common factors, (i) by use of cross-section averages of model variables, CA (following Pesaran, 2006), or (ii) by estimating them using principal component analysis, PCA (following Bai, 2009). Since the former has precedence in an unpublished extension to Boneva and Linton (2017), we adopt the CA augmentation approach in combination with the treatment effects estimator by Chan and Kwok (2022): we construct the cross-section averages from the sample of ‘never treated’ countries (in our case: countries which remained autocratic throughout the sample period) and provide an equivalent test to these authors’ Alpha test for weak parallel trends.¹ Our estimation equation is a treatment equation for those countries which did experience regime change during the sample period. The empirical implementation is via Poisson pseudo-maximum-likelihood (PPML) at the pair level to address concerns over heteroskedastic errors and zero trade flows, in line with the existing best practice in this literature spurred by Santos Silva and Tenreyro (2006).²

Our empirical investigation relates to the important question of the effect of institutions, broadly defined, on economic activity. Until very recently the literature on the democracy-growth nexus was marred by the absence of causal evidence for a ‘democratic dividend’, which in the face of rampant economic progress in autocratic China in particular led to sceptical views among academics and policymakers over the relevance of democracy for prosperity. Existing empirical research wishing to address this question within the gravity model framework had to adopt questionable workarounds, for instance, either defining the presence or absence of democracy at the country pair-level or fudging the high-dimensional fixed effects approach dictated by the structural gravity model straitjacket. In our analysis, we employ rich hierarchical data on liberal democracy and its constituent components (electoral democracy and the liberal component) from V-Dem (Varieties of Democracy) to investigate the impact of democratic regime change on bilateral trade flows. Adopting the heterogeneous gravity approach in combination with novel heterogeneous treatment effect models (Chan and Kwok, 2022) allows us to make a strong case for a credibly causal effect isolated by our analysis. We relate the analysis of the democratic regime change effect on trade to the time spent in democracy presenting predictions from multivariate running line regressions and exploring heterogeneities over time (Cold War, post-Cold War eras) and across countries (adopting separate treatment and control samples for countries with, respectively, favourable and unfavourable deep determinants of comparative development related to geography, legal origin, culture, and history).

We find that the economic effects of democratic regime change are sizeable: the average country

¹Like all difference-in-differences estimators relying on the assumption of parallel trends before treatment the Chan and Kwok (2022) estimator relies on the weaker assumption of equal average factor loadings on the common factors between treated and control samples — see Section 3 for more details.

²We still need a further correction to address the practical issues of country pairs where the exporter or importer experience democratic regime change but no trade ever takes place between the two countries.

experiences a 40% rise in exports after twenty years of democracy, an effect which is economically large and stable over the long run. The long-run effect shows virtually no differences across *importer* regimes (always democracy, always autocracy, democratisers). We conduct several robustness checks including alternative definitions of democratic regime change but the patterns remain qualitatively the same. We see few differences in the trajectories of a trade effect over the years in democracy between the building blocks of liberal democracy — electoral democracy/polyarchy and the liberal component — yet the latter’s long-run effects are distinctly smaller than the former’s. Dividing the data into two periods for the Cold War and post-Cold War eras shows that democratic regime change had no trade effect in the former era, whereas the patterns in the analysis of 1990-2014 point to relatively stable but substantially lower trade effects, at around 20% after ten years in democracy. The division into Cold War and post-Cold War eras also substantially influences our analysis of deep determinants of comparative development and unequal benefits of regime change: although the analysis of the full sample years suggests that countries with unfavourable deep determinants (higher disease environment and unfavourable climate, distant culture, colonial experience) saw lower trade effects from regime change than their peers, the analysis of the 1990-2014 sample sees this effect substantially increased: once ideology no longer dominated economics, the causal effect of regime change on exports in countries with unfavourable geography, French legal origin or distant cultural traits is substantially lower.

We make three contributions to the literature: first, we introduce a simple but powerful new empirical implementation of the gravity model of trade and other flows which allows for the identification of monadic variables. Second, we study the relationship between country-specific democratic regime change and bilateral trade flows, contributing to the investigation of underlying transmission mechanisms of the large democracy-growth effect. Third, we link our heterogeneity analysis of unequal benefits from democratic regime change to the work on ‘deep determinants of comparative development’ to demonstrate that such deep structural characteristics prevail and can substantially diminish the large positive economic effect of democratic regime change.

The remainder of this paper is structured as follows: in the next section we review the existing literature on the gravity model, on democracy and growth, the intersection between these literatures, and an update on recent macro panel econometric contributions. Section 3 introduces our heterogeneous gravity model approach, and Section 4 covers the data. Our main results are reported in Section 5, including our investigation of secular and cross-country heterogeneity. Section 6 concludes.

2 Background

Gravity The positive effects of trade and foreign direct investment (FDI) on economic development have been well documented (e.g. Bruno et al., 2018; Feyrer 2019). They generate new activities,

facilitate technology transfer, and contribute to static and dynamic efficiency gains. These flows of goods and capital have thus been recognised as forces that ought to be encouraged by governments to make the most of the opportunities offered by globalisation, and, in response, a large literature has investigated their determinants. The empirical vehicle of choice in this regard is the gravity model of bilateral trade flows, which is also applied in the study of bilateral FDI (e.g., Bergstrand and Egger, 2007; Head and Ries, 2008), equity (e.g., Portes and Rey, 2005) and migration flows (e.g., Bertoli and Fernández-Huertas Moraga, 2013).

Over the past six decades since the first application to international trade by Tinbergen (1962) the literature analysing trade flows between countries has experienced a tremendous amount of progress: first, researchers have been able to build theoretical foundations for the empirical regularity of a 'gravity' force in international trade flows, which, in the Newtonian fashion, relates the volume of trade between countries to the size of their respective economies, and the presence/absence of trade 'frictions' (e.g. their geographical distance, regional trade agreements, colonial link, or common language). The theoretical foundations were led by Anderson (1979), Bergstrand (1985), Eaton and Kortum (2002), and Anderson and van Wincoop (2003), and the gravity model can be derived from a wide class of microeconomic principles, see for example, Arkolakis et al. (2012), Costinot and Rodriguez-Clare (2014), and Allen et al. (2020). Second, empirical researchers recognised that trade between country pairs cannot be viewed in isolation, but that in line with the theory, the entire network structure of trade flows (or their absence between certain country pairs) needs to be considered (a concept referred to as 'multilateral resistances', MRTs, see Anderson and van Wincoop, 2003) to avoid an omitted variables bias.

By now, taking on board the discussed concerns and developments, a consensus in the literature with regards to the detailed 'form' of the empirical specification to be estimated (interactions between exporter and importer fixed effects and year fixed effects; fixed effects for country pairs) as well as the empirical estimator to be adopted (Poisson Pseudo-Maximum-Likelihood, PPML, see Santos Silva and Tenreyro, 2006) emerged. These developments led to an empirical gravity equation that has remarkable predictive power (for recent surveys see Head and Mayer, 2014; Yotov et al., 2016), which also explains its widespread application to explain trade flows and the impact of the determinants of trade flows in thousands of academic papers. Importantly, the gravity equation is heavily adopted in policy analysis. These innovations naturally extend to the literature on FDI, migration and remittance flows, which are also modelled within the gravity empirical framework.

An alternative reading of these developments is that in order to get close to the theory, the questions addressed by empirical work in trade research have become extremely narrow. Once the battery of necessary fixed effects has been added to the pooled gravity regression, the signing of new regional trade agreements (and a small range of other dyadic variables, i.e. those varying at

the country-pair level) is about the only (policy) measure left to analyse. Is this truly a question of first-order significance? Maybe yes, maybe no. However, one should think that ‘the rest’ — all those monadic variables — also matter a great deal: for example, non-discriminatory trade policies, institutions, or macroeconomic policies. Since the current inferential ‘technology’ is very rigid, structural gravity researchers have largely turned away from considering some of the big questions about the determinants of trade flows. The effects of country-specific characteristics or policies on international trade and FDI flows are quite commonly of great interest to researchers and policymakers but as such not identifiable in a well-specified structural gravity equation. Existing work either uses a-theoretical frameworks, constructs bilateral variables from country-specific ones in an *ad hoc* manner, or relies on the identification of the differential effect between inter-national and intra-national trade flows (the latter difficult to measure) to estimate the effects of country-specific variables.

Democracy and Growth The empirical confirmation that democratisation *in the average country* has a large, positive, and causal effect on economic prosperity is surprisingly recent (Madsen et al, 2015; Acemoglu et al, 2019; Eberhardt, 2022).³ However, there is widespread scepticism among economists and political scientists about this finding. First, there is the suggestion that allowing voters to remove an incumbent government with the power of the electoral process would drive up consumption and reduce the investment rate to the detriment of economic growth (see Baum and Lake, 2003, 334f for this argument). Studying China or the East Asian ‘Tiger’ economies suggests that democracy is not a necessary condition for economic prosperity, but this form of cherry-picking of individual success stories often ignores the established link between autocracy and large variation in economic performance (Persson and Tabellini, 2009; Knutsen, 2012). Acemoglu et al. (2019) recently provided convincing evidence of a large causal economic effect of regime change, yet several key challenges remain. It is, for instance, commonly acknowledged that the magnitude of economic benefits will vary across countries (Gerring et al., 2005; Boese-Schlosser and Eberhardt, 2023), though there is little research on what may explain such differences. Furthermore, the channels of transmission from improved political institutions to economic prosperity have not been explored in systematic detail. Both aspects, the heterogeneity, and channels in the form of trade effects, are studied in the present paper.

Democracy and Trade In contrast to the literature on economic development, the existing empirical research on democracy and trade is surprisingly sparse (Anderson and Marcoullier, 2002; Baier and Bergstrand, 2007; Yu, 2010; Beverelli et al, 2024) and is subject to our general straitjacket criticism of empirical gravity research. Theoretical considerations regarding a democracy-trade nexus build

³A detailed discussion of the literature going back to Barro (1996) and earlier political science contributions is provided in Boese-Schlosser and Eberhardt (2023).

on institutional notions of property rights protection and fair competition on the exporter side, implying that exporting firms are enabled to develop higher-quality products (in part due to higher R&D expenditure), which then naturally create a global demand (Yu, 2010). On the importer side, improving democratic institutions is associated with a shift to pro-trade policies in developing countries, where power shifts away from the protectionist political and economic elite (Milner and Kubota, 2005) and the reverse in advanced economies (O'Rourke and Taylor, 2006), echoing the widespread suggestion that the median voter is poor and favours consumption over investment (e.g. Przeworski et al, 2000), hence suggesting an ambiguous effect of democracy on trade. It should be noted that the latter argument of a diversion from investment to consumption is similarly applied in the democracy-growth context (see Baum and Lake, 2003, for details) yet the empirical literature has found resoundingly large and positive causal effects (Acemoglu et al, 2019; Eberhardt, 2022; Boese-Schlosser and Eberhardt, 2024).

Macro panel econometrics The empirical methodology we seek to develop and exploit extends the existing econometric literature on the multifactor error structure (e.g. Pesaran, 2006; Bai, 2009) from the linear to the generalised linear/non-linear model (Boneva and Linton, 2017). In many empirical macro applications, ranging from forecasting (Stock and Watson, 2002) to cross-country growth empirics (Eberhardt et al, 2013; Eberhardt and Presbitero, 2015; Boese-Schlosser and Eberhardt, 2023, 2024), common factors are employed to reduce the vast number of potential covariates and/or to capture unobserved time-varying heterogeneity. The underlying, principal idea is that macroeconomic variables such as GDP, inflation, or trade flows, are driven by a small set of unobserved common factors, with country (or country-pair) specific parameters, so-called factor loadings. The factors and loadings are unknown, but in empirical applications can be estimated or proxied. since in most applications their magnitudes are not of particular interest, it is merely recognised that they need to be accounted for to avoid bias in the estimated parameters of interest — see Eberhardt and Teal (2011) for an introduction. There are two ways of achieving this aim: (i) including cross-section averages of all model variables in the empirical equation, following Pesaran (2006); or (ii) estimating the common factors via Principal Component Analysis, following Bai (2009). Unpublished work in an extension to Boneva and Linton (2017) provides suggestions that the cross-section average augmentation for the Probit-CCE estimator they developed in the latter paper is also valid for the generalised linear model, including a Poisson Maximum Likelihood implementation.

Building on this extension we provide a further innovation: in the context of binary regressors (democratic regime change) we combine the heterogeneous gravity model with recent contributions to heterogeneous difference-in-difference estimation (Chan and Kwok, 2022): modelling a 'treatment' as an endogenous, binary event, we include cross-section averages computed from a control sample

of countries (here: exporters or importers) which did not experience the treatment.⁴ This empirical implementation furthermore deals with the potential for non-parallel pre-intervention trends as well as for nonstationary variable series and processes – see Boese-Schlösser and Eberhardt (2023, 2024) for applications using standard rather than dyadic panel data.

3 Methodology

In this section, we develop a novel empirical implementation to capture monadic variables in a factor-augmented heterogeneous gravity model. The first two sub-sections are a recap of the modern structural gravity empirics, before section 3.3 introduces heterogeneous gravity. Our combination of the PPML-CCE and recent heterogeneous difference-in-differences estimators is presented thereafter. The details for an equivalent diagnostic test of the standard parallel trend assumption are provided in section 3.5. A novel way of presenting treatment effects when treatment length differs substantially, introduced in Boese-Schlösser and Eberhardt (2024), is discussed in section 3.6.

3.1 The Gravity Equation

We assume a gravity relationship in the panel in which bilateral trade of exporter i to destination market j at time t is given by⁵

$$X_{ijt} = \frac{Y_{it}}{\Omega_{it}} \frac{X_{jt}}{\Phi_{jt}} \phi_{ijt} \quad \text{where } 0 \leq \phi_{ijt} \leq 1. \quad (1)$$

X_{ijt} is a trade flow from an exporter to a destination market, Y_{it} is the value of production for the exporter and X_{jt} the value of expenditure in the destination market j on all source countries — the latter two are typically proxied by GDP in the exporter and destination markets, respectively.⁶ ϕ_{ijt} captures the ‘bilateral accessibility’ for destination j and exporter i : this contains trade costs between the two markets and any variable which may affect these, including time-variant and invariant, observed and unobserved factors.

A major development in gravity modelling over the past decade following the seminal contribution by Anderson and Van Wincoop (2003) is the recognition that the conditional trade between destination j and exporter i (the conditions being the ‘bilateral accessibility’) cannot be viewed in isolation from the set of opportunities open to importer j in sourcing goods from exporters other than i and the

⁴Since (i) trade flows are not symmetric (some country i may export to country j but the reverse may not be the case); and (ii) the heterogeneous treatment effects model limits regressions to those countries which did experience regime change, the cross-section averages for importers which never democratised and exporters which never democratised are not identical.

⁵This exposition builds on the discussion in Santos Silva and Tenreiro (2006), and Yotov et al. (2016) for the gravity model, and Cameron and Trivedi (1998) for Poisson regression.

⁶If X_{ij} is merchandise trade then theory-consistency dictates Y_i to be gross production of traded goods (not simply value-added/GDP) and X_j the apparent consumption of goods, production plus imports minus exports (Head and Mayer, 2014).

relative access exporter i has to destinations other than j .⁷ The multilateral resistance variables for each actor in the exchange of goods are defined in terms of the bilateral accessibility-weighted exporter capabilities and importer characteristics respectively: exporter $i = 1, \dots, N - 1$ and for importer $j = 1, \dots, N - 1$, and $i \neq j$ let

$$\Omega_{it} = \sum_{\ell=-i} \frac{\phi_{i\ell t} X_{\ell t}}{\Phi_{\ell t}} \quad \Phi_{jt} = \sum_{\ell=-j} \frac{\phi_{\ell j t} Y_{\ell t}}{\Omega_{\ell t}} \quad (2)$$

where $-j$ and $-i$ signify that these magnitudes are not defined in reflexive terms and thus exclude destination j and exporter i from the respective MRTs of the trading relationship between these two markets.

For our derivation of the empirical gravity model, we assume a stochastic version of equation (1)

$$X_{ijt} = \frac{Y_{it}}{\Omega_{it}} \frac{X_{jt}}{\Phi_{jt}} \phi_{ijt} \eta_{ijt} \quad (3)$$

where η_{ijt} is an error factor with $E[\eta_{ijt}|Y_{it}, \Omega_{it}, X_{jt}, \Phi_{jt}, \phi_{ijt}] = 1$.

A very general empirical equivalent to equation (3) allows for flexible unknown parameters on the observable mass and accessibility variables:

$$X_{ijt} = \exp[\beta_{ijt}^i \ln(Y)_{it} + \beta_{ijt}^n \ln(X)_{jt} + \gamma_{ijt} \ln(\phi)_{ijt} + \ln(\Omega)_{it} + \ln(\Phi)_{jt}] \eta_{ijt}, \quad (4)$$

where superscripts are used to identify the coefficient of exporter versus importer GDP/expenditure. This specification is the most general empirical model possible where all unknown parameters on observable variables $(\beta^i, \beta^j, \gamma)$ vary at the pair level and over time. We demonstrate how this relates to the models employed in the existing literature by adopting parameter restrictions.

3.2 Model Restrictions and Pooled Estimation

A first restriction, adopted in the vast majority of studies, is to assume the gravity model estimates are fixed *across time*:

$$X_{ijt} = \exp[\beta_{ij}^i \ln(Y)_{it} + \beta_{ij}^n \ln(X)_{jt} + \gamma_{ij} \ln(\phi)_{ijt} + \ln(\Omega)_{it} + \ln(\Phi)_{jt}] \eta_{ijt}, \quad (5)$$

A notable exception here is the study by Klasing, Milionis, and Zymek (2015) who allow for time-variation in three distinct regimes over their long panel from 1870 to 2005. In our analysis of the post-WWII period we first follow the bulk of the literature and estimate policy effects which are

⁷This network of dependencies is formalised by econometricians as the deviation from the assumption of ‘cross-section weak dependence’ (Andrews, 2005; Chudik, Pesaran, and Tosetti, 2011; Chudik and Pesaran, 2015b).

specified as time-invariant; later on we *partly* relax the assumption of fixed parameters over time by adopting 20-year rolling regression windows.

Conventionally, further restrictions in the panel gravity literature are to assume common parameters on the observable variables ($\beta_{ij}^i = \beta^i$, $\beta_{ij}^n = \beta^n$, $\gamma_{ij} = \gamma$). Pairwise fixed effects (δ_{ij}) are added to capture trade policy endogeneity (Baier and Bergstrand, 2007). Additionally including exporter-time (ω_{it}) and importer-time (ψ_{jt}) fixed effects can capture the MRTs (Hummels, 2001; Anderson and van Wincoop, 2003; Feenstra, 2004) — with implications for β^i and β^j (see below).

$$X_{ijt} = \exp[\beta^i \ln(Y)_{it} + \beta^j \ln(X)_{jt} + \gamma \ln(\phi)_{ijt} + \delta_{ij} + \omega_{it} + \psi_{jt}] \eta_{ijt}. \quad (6)$$

Thus δ_{ij} , ϕ_{jt} and ω_{it} are the unknown parameters estimated on the various fixed effects in the reduced-form panel gravity model.

3.3 Heterogeneous Parameter Estimation

A practical difficulty arises if a more flexible specification for the observable variables such as that laid out in equation (5) on the one hand (β_{ij}^i , β_{ij}^n , γ_{ij}), and the aforementioned recommended practice to capture MRTs on the other are to be combined: for pairwise heterogeneity in the economic mass and policy variable parameters it is most convenient to estimate equation (5) *separately* for each country pair, which trivially also addresses policy endogeneity since δ_{ij} is captured by an intercept in this pair-level time series regression. However, we cannot include fixed effects for *all exporters and all importers* in an equation for a single importer-exporter pair, let alone interacted with time dummies. Existing studies in the literature which allow for trade policy heterogeneity employ interaction effects (e.g. Baier, Bergstrand and Clance, 2018) or maintain a set of fixed effects in a PPML model but use pair-specific trade policy dummies (Baier, Yotov and Zylkin, 2019). An alternative approach is to draw on recent insights from the panel time series literature (e.g. Pesaran, 2006; Bai, 2009) and to employ a multi-factor error structure to capture the unobserved MRTs.

First, we bring the error factor on the inside of the exponential function to capture all unobservables u_{ijt} :

$$\begin{aligned} u_{ijt} &= \delta_{ij} + \omega_{it} + \psi_{jt} + \ln(\eta)_{ijt} \\ &\equiv \delta_{ij} + \omega_{it} + \psi_{jt} + \varepsilon_{ijt}. \end{aligned} \quad (7)$$

Next, the dimensionality problem of dealing with a large number of unknown parameters (the number of ω_{it} and ϕ_{jt} swiftly add up to thousands of directional-time dummies) in a country-pair equation with about 70 post-WWII time series observations can be solved by imposing more structure on these

unobservables. In the macro panel econometric literature it is widely acknowledged that a small number of common factors with heterogeneous factor loadings can represent large datasets of macro variables. For instance, in the forecasting literature Stock and Watson (2002) have shown that 149 macroeconomic time series can be reduced to two or three principal components. Furthermore, Bai (2009) discusses several macro-, microeconomic and finance applications where common factors can be employed to model unobserved time-varying heterogeneity in a tractable way.⁸

In the case at hand, we posit that the economic mass and accessibility variables along with the trade flows and MRTs are all driven by a small number of common factors with heterogeneous factor loadings across country pairs. Thus, we argue that a small number of unobserved common factors $\mathbf{f}_t$, each with country pair-specific factor loadings φ_{ij} , can account for the evolution of trade flows, GDP, etc. In the notation introduced above:

$$u_{ijt} = \delta_{ij} + \omega_{it} + \psi_{jt} + \varepsilon_{ijt} \quad (8)$$

$$\approx \delta_{ij} + \varphi'_{ij}\mathbf{f}_t + \varepsilon_{ijt}, \quad (9)$$

where an ‘approximate factor structure’ is represented by a set of common factors $\mathbf{f}$, and the associated factor loadings φ .

The insight gained in the recent panel time series literature from this setup in the *linear* regression case is that the unobserved common factors can be captured by observables, either via principal component analysis (Bai, 2009) or using cross-section averages of the dependent and independent variables (Pesaran, 2006). We briefly develop the latter approach (the ‘common correlated effects’ or CCE estimator) and provide a mathematical indication of the intuition at play — this is for ease of illustration since it will only be a small step in terms of implementation from the linear model to a generalised linear one (Boneva and Linton, 2017).

For simplicity we assume a double-index of t for the time series and i for the cross-section — we can think of the latter as a placeholder for the country pair as the unit of analysis like in the gravity model, i.e. $i \equiv ij$. Let

$$y_{it} = \alpha_i + \beta_i x_{it} + u_{it} \quad u_{it} = \lambda_i \mathbf{f}_t + \varepsilon_{it} \quad (10)$$

$$x_{it} = \delta_i + \phi_i \mathbf{f}_t + \varrho_i g_t + e_{it}, \quad (11)$$

where ε and e are white noise processes. This setup indicates that the regressor x is driven by the same common factor $\mathbf{f}$ as the dependent variable y , albeit with different parameters.⁹ In addition there

⁸The multifactor error structure has been applied to capture country-specific time-varying total factor productivity (Eberhardt and Presbitero, 2015), time-varying absorptive capacity (De Visscher et al, 2020), or knowledge spillovers in the analysis of sector-level production functions augmented with sectoral R&D stock (Eberhardt, Helmers, and Strauss, 2013).

⁹In a setup with multiple factors we can instead assume that only a subset of $\mathbf{f}$ ‘overlaps’ between the two equations.

are some factors g which only drive x but not y . This setup is standard in the macro panel literature and we refer to the studies in footnote 8 for details on factor evolution, parameter distributions, etc. It is clear from equations (10) and (11) that x is endogenous and that failing to account for the presence of the unobserved common factors will lead to omitted variable bias.¹⁰

Pesaran's (2006) approach posits that the unobserved common factor f can be captured by the cross-section averages of y and x provided the cross-section dimension of the panel is not too small.¹¹ A simple algebraic derivation can provide intuition for the mechanism at work: take the cross-section average of equation (10) and solve it for the common factor f :

$$\bar{y}_t = \bar{\alpha} + \bar{\beta}\bar{x}_t + \bar{\lambda}f_t \Leftrightarrow f_t = \bar{\lambda}^{-1}(\bar{y}_t - \bar{\alpha} - \bar{\beta}\bar{x}_t), \quad (14)$$

where the bars indicate cross-section averages and the error term disappears since $\bar{\varepsilon} = 0$ by assumption. Next, plug the expression for f back into the original equation (10)

$$y_{it} = \alpha_i + \beta_i x_{it} + \lambda_i \bar{\lambda}^{-1}(\bar{y}_t - \bar{\alpha} - \bar{\beta}\bar{x}_t) + \varepsilon_{it} \quad (15)$$

$$= \alpha_i - \lambda_i \bar{\lambda}^{-1} \bar{\alpha} + \beta_i x_{it} + \lambda_i \bar{\lambda}^{-1} \bar{y}_t - \lambda_i \bar{\lambda}^{-1} \bar{\beta} \bar{x}_t + \varepsilon_{it},$$

$$y_{it} = \varpi_i + \beta_i x_{it} + \zeta_i \bar{y}_t + \vartheta_i \bar{x}_t + \varepsilon_{it}, \quad (16)$$

where we reparameterize in the last line. Thus the unobserved common factor f can be captured by the cross-section averages of y and x , while the heterogeneous impact of f across i can be captured by estimating equation (16) separately for each panel member — the principle extends to multiple factors. A Mean Group estimator following Pesaran and Smith (1995) captures the central tendency of the panel and provides a convenient comparison with alternative pooled empirical models:

$$\hat{\beta}^{MG} = N^{-1} \sum_{i=1}^N \hat{\beta}_i \quad (17)$$

Inference for the Mean Group estimates is based on a simple nonparametric variance estimator

¹⁰Solving the x equation for f and plugging this into the y equation yields

$$y_{it} = \alpha_i + \beta_i x_{it} + \lambda_i \phi_i^{-1}(x_{it} - \delta_i - \psi_i g_t - e_{it}) + \varepsilon_{it} \quad (12)$$

$$\begin{aligned} &= \alpha_i - \lambda_i \phi_i^{-1} \delta_i + (\beta_i + \lambda_i \phi_i^{-1}) x_{it} - \lambda_i \phi_i^{-1} \psi_i g_t - \lambda_i \phi_i^{-1} e_{it} + \varepsilon_{it} \\ &= \eta_i + \theta_i x_{it} + \nu_{it}, \end{aligned} \quad (13)$$

where in the final line we reparameterise. Crucially, unless $\lambda_i \phi_i^{-1} = 0$ we can see that β_i is unidentified. The asymptotic bias will be a function of the (relative) 'strength' of the factors in their impact on y and x in panel member i .

¹¹Additional robustness of this approach to nonstationary factors, structural breaks, and additional spatial dependence, among other aspects, is discussed in Kapetanios, Pesaran, and Yamagata (2011), Chudik, Pesaran and Tosetti (2011), Chudik and Pesaran (2015a) and Westerlund (2019).

(Pesaran and Smith, 1995; Pesaran, 2006):¹²

$$\hat{\Omega}^{MG} = \frac{1}{N-1} \sum_{i=1}^N (\hat{\beta}_i - \hat{\beta}^{MG})(\hat{\beta}_i - \hat{\beta}^{MG})'. \quad (18)$$

Boneva and Linton (2017) extend the above setup from the linear model to a setting where the outcome variable is discrete. They show that the CCE approach can be applied to a probit model by including only the cross-section averages of the observed regressors: under the assumption that the unobserved factors are contained in the span of the cross-section averages of the regressors they derive asymptotic results for the large T , large N case as well as the consistency and asymptotic normality of the Mean Group estimator of the individual-specific estimates.

The same principle can be applied to a generalised linear model: in our case, we begin by assuming the exponential mean function incorporating a multi-factor error structure¹³

$$E[y_{it}|x_{it}, \mathbf{f}_t] = \mu_{it} = \exp[\alpha_i + \beta_i x_{it} + \boldsymbol{\lambda}'_i \mathbf{f}_t]. \quad (19)$$

The Poisson Maximum Likelihood (PML) estimator assumes that the distribution of y given x (and the factors) is Poisson (i.e. a count variable), but it is widely recognised that the data generating process is not required to be Poisson for this estimator to be consistent (Cameron and Trivedi, 1998; Santos Silva and Tenreyro, 2006), in which case it is referred to as a Poisson Pseudo Maximum Likelihood (PPML) estimator. Our implementation uses the estimator in the single time series, namely of the pair-wise gravity equation with exponential mean function, where the common factors are replaced by cross-section averages of the regressors, which we refer to as PPML-CCE:

$$\begin{aligned} \mu_{ijt} = & \exp[\alpha_{ij} + \beta_{ij}^i \ln(Y)_{it} + \beta_{ij}^n \ln(X)_{jt} + \gamma_{ij} \ln(\phi)_{ijt} \\ & + \delta_{ij} \overline{\ln(Y)}_t + \kappa_{ij} \overline{\ln(\phi)}_t] + \epsilon_{ijt} \quad \forall ij, j \neq i. \end{aligned} \quad (20)$$

In our case, the accessibility term (any variable that may affect bilateral trade costs) is replaced by dummies for democratic regime change in i and j respectively along with dyadic dummies for free trade agreements and currency unions (in robustness checks we omit the latter two since they may be viewed as ‘bad controls’).¹⁴ In practice, where exporter and destination mass are proxied using their respective GDP, the cross-section averages are identical, $\overline{\ln(Y)}_t = \overline{\ln(X)}_t$. In our difference-in-differences implementation below these come from distinct control samples and are thus separately

¹²In practice we follow the standard in the literature and employ robust means (Hamilton, 1992) to reduce the effect of outliers.

¹³We thank Lena Boneva and Oliver Linton for sharing the rough derivations for a CCE-augmentation in this more general case.

¹⁴We know that the dependent variable, conditional on the regressors, can be heteroskedastic, and serially correlated and that the PPML estimator maintains its consistency in the presence of serial correlation if the exponential conditional mean is correctly specified (Cameron and Trivedi, 1998: 226) — the latter is the requirement for any PML or PPML estimator.

identified.

Our empirical estimates below are thus based on cross-section average-augmented time series PPML regressions at the country-pair level which are subsequently averaged following the Mean Group principle; inferential statistics for the PPML-CCE Mean Group estimate are computed using the variance estimator in (18): Boneva and Linton (2017) have shown the simple nonparametric variance estimator still applies in the generalised linear model setup.

Estimation of the PPML-CCE model can be seen as an improvement on current practice for two reasons. First, the presence of common factors, with heterogeneous loadings across country pairs, offers a flexible way to account not only for the MRTs, but also so-called globalisation effects, and other forms of spatial dependence. Second, using common factors as proxies for spatial dependence allows the estimation of country-specific (monadic) variables — thus in addition to dyadic determinants such as those contained in $\ln(\phi)_{ijt}$, we can also introduce country-specific variables of interest.

3.4 A Difference-in-Differences Gravity Model

Our exposition so far has focused on heterogeneous gravity without giving too much thought to the empirical practices in the democracy-growth literature, where economists favour the use of *binary* indicators for democracy versus autocracy (Papaioannou and Siourounis, 2008; Acemoglu et al, 2019; Eberhardt, 2022; Boese-Schlösser and Eberhardt, 2023, 2024). We now introduce a combination of the PPML-CCE model with a factor-augmented difference-in-differences estimator following Chan and Kwok (2022): let D_{it} be a dummy for democratic regime change equal to 0 when country i is in autocracy and 1 if they are in democracy — we refer to the shift from 0 to 1 as ‘treatment’. The Chan and Kwok (2022) principal component DID (PCDID) estimator estimates the causal effect of a treatment in country-specific regressions of treated countries only. While this appears to capture only the first of the difference-in-differences, the comparison with the control sample (in our case those countries which never experienced democratic regime change) is achieved by the inclusion of unobserved common factor proxies extracted from the control sample of never-democratisers. Crucially, this heterogeneous difference-in-differences estimator does not require the ‘parallel trends’ assumption to hold, but relies on a much weaker assumption of ‘weak parallel trends’ between treated and control samples — we provide more details in the next section where we develop a suitable test in the following subsection.

In the context of democratic regime change and the PPML-CCE gravity model of trade flows, we make some adjustments to the way the Chan and Kwok (2022) PCDID is implemented: first, we do not *estimate* the unobserved common factors from auxiliary regressions in the control sample of never-democratisers. Extracting several principal components via PCA from the residuals of a

PPML regression is conceptually difficult (given that these are not linear models) and in practice raises concerns over how many estimated factors to include. Instead, we use cross-section averages¹⁵ of GDP and population in line with Boneva and Linton (2017). Second, these cross-section averages are constructed from the respective control samples of (i) exporters and (ii) destinations: since the trade flow panel is not symmetric (i may be exporting to j but j does not export to i) these two cross-section averages are not identical.

Our PPML-CCE-DID estimator is then implemented as follows:

$$\begin{aligned} \mu_{ijt} = & \exp[\alpha_{ij} + \gamma_{ij}^1 \text{FTA}_{ijt} + \gamma_{ij}^2 \text{Common Currency}_{ijt} + \theta_{ij} D_{it} + \eta_{ij} D_{jt} \\ & + \delta_{ij}^i \overline{\ln(Y)}_t^i + \kappa_{ij}^i \overline{\ln(Pop)}_t^i + \delta_{ij}^j \overline{\ln(Y)}_t^j + \kappa_{ij}^j \overline{\ln(Pop)}_t^j + \epsilon_{ijt}] \quad \forall ij, j \neq i. \end{aligned} \quad (21)$$

We have pair fixed effect (α_{ij}) and the familiar dyadic variables for FTA and common currency union. The two democracy dummies (D_{it} for exporter i and D_{jt} for destination j) are accompanied by two sets of cross-section averages, which are constructed from the exporter i and destination j control samples of countries which never democratised, respectively. We do not include cross-section averages of the indicator variables in the first line of equation (21) since this practice is likely to induce severe bias in the estimation equation (Juodis et al, 2021). Our coefficient of interest is θ_{ij} , which is estimated (separately) in each trade flow equation of country i with each of its trade partners j — as is standard in the heterogeneous DID literature, we estimate equation (21) only for equations where either country i or country j experienced regime change.

We cannot estimate the democracy effect in country-pairs where trade flows remained zero throughout the sample period, whether they experienced regime change or not: this is necessitated by our heterogenous regression implementation. We do however include the GDP and population variables for these country pairs in the computation of the cross-section averages. More worryingly, excluding country pairs where the exporter or importer did experience regime change but nevertheless, trade between the pair never materialised, will likely induce an upward (selection) bias in our results. We therefore adjust the model to include cross-section averages from those country pairs where the exporter or importer country did experience regime change but where trade flows remained zero. Our final empirical implementation is thus:

$$\begin{aligned} \mu_{ijt} = & \exp[\alpha_{ij} + \gamma_{ij}^1 \text{FTA}_{ijt} + \gamma_{ij}^2 \text{Common Currency}_{ijt} + \theta_{ij} D_{it} + \eta_{ij} D_{jt} \\ & + \delta_{ij}^i \overline{\ln(Y)}_t^i + \kappa_{ij}^i \overline{\ln(Pop)}_t^i + \delta_{ij}^j \overline{\ln(Y)}_t^j + \kappa_{ij}^j \overline{\ln(Pop)}_t^j \\ & + \delta_{ij}^{i0} \overline{\ln(Y)}_t^{i0} + \kappa_{ij}^{i0} \overline{\ln(Pop)}_t^{i0} + \delta_{ij}^{j0} \overline{\ln(Y)}_t^{j0} + \kappa_{ij}^{j0} \overline{\ln(Pop)}_t^{j0} + \epsilon_{ijt}] \quad \forall ij, j \neq i, \end{aligned} \quad (22)$$

¹⁵Eberhardt (2022) adopts the same strategy in a standard (monadic) panel dataset to study democracy and growth.

where the superscript $i0$ ($j0$) indicates the cross-section averages or associated parameters from exporter (destination) countries which did experience regime change but where trade flows between i and j remained zero throughout the sample period. This specification is even more demanding on the data, with 13 parameters to be estimated in each country-pair regression. A variation on the above is a ‘plain vanilla’ model which excludes the additional controls (in our case pairwise FTA and Currency Union dummies) since these could represent bad controls (in the sense of Angrist and Pischke, 2009).

3.5 Diagnostic Testing

The reliability of every difference-in-differences result lives and dies by the validity of the ‘parallel trend’ assumption: if the soon-to-be-treated unit was already on a different trajectory to the control unit(s) before treatment (due to some form of selection into treatment), then the standard difference-in-differences estimates are biased. In the Chan and Kwok (2022) PCDID case the estimator allows for non-parallel trends across panel units, most importantly between those in the treated and those in the control samples. Nevertheless, the estimator requires a form of the parallel trend test for validity, albeit in a much weaker form: the average factor loadings on the common factors (extracted from the control sample) in the treatment equation have to equal those in the control equation. In practice, this implies that the estimated parameters on the factor proxies in the treatment equation statistically should not differ from unity.

The intuition of this test is to ask whether the information we extract from the control sample is equally relevant in the treatment sample: imagine a world where democratic regime change would only happen to the richest countries in the world, whereas the sample of countries which always remained autocratic was made up of the poorest economies. The ‘information’ contained in the factor proxies constructed from the poor country control sample is then likely to be quite uninformative about the unobservables driving outcomes in the rich country treatment sample: countries at different ends of the income scale have different economic structures, financial systems, physical and human capital stocks, etc. The unobservables (or hard to measure factors) driving, say, income in Mali (rainfall, absence of civil unrest) are likely to be very different from those driving income in the United States (innovation, geopolitics) — we’d be comparing apples and oranges. Passing the Alpha test assures us that the unobservables are *on average the same* across treatment and control samples, that we are comparing apples with apples, or oranges with oranges.

In comparison to the Chan and Kwok (2022) Alpha test, in the present setup using cross-section averages rather than common factors estimated from auxiliary regression residuals, we cannot use their testing strategy, which employs the cross-section average of the residuals from the auxiliary regressions in the control sample. We devise a ‘Pseudo Alpha’ test for the heterogeneous trade flow

regressions augmented with cross-section averages by testing whether the factor loadings between control sample and treated sample are on average the same, meaning that the average coefficients $\hat{\delta}^i$, $\hat{\delta}^j$, $\hat{\kappa}^i$ and $\hat{\kappa}^j$ in equation (21) will jointly be equal to 1. If the underlying null of equal average factor loadings between treatment and control samples is rejected, we have to be concerned about the treatment regression being misspecified and hence arriving at biased treatment estimates.

3.6 Treatment Effects and Years in Treatment

A pooled difference-in-differences estimator provides the average treatment effect on the treated (ATET). Although event studies can give a sense of the evolution of the treatment effect, this choice is by necessity: a pooled model can only report the average across the entire treatment sample — the nature of this average has recently been subject to a great deal of research since the potential for negative weights was raised by de Chaisemartin and d'Haultfoeuille (2020).

In the present case, we follow a practice introduced in Boese-Schlösser and Eberhardt (2024) which employs multivariate running line regression (Royston & Cox 2005) to enable us to flexibly present the relationship between the treatment effect and the years a country (exporter) spent in democracy: if democratic regime change affects countries' innovative capacity, then this may represent an endogenous growth engine which creates a permanent change in the economic growth rate, rather than a one-off levels effect. Our mode of presentation does not rule out one or the other but simply enables us to study the cross-country trajectory of the democracy-trade relationship conditional on the number of years spent in democracy. In addition, this approach enables us to control for differential data availability (start year of the pair-wise regression sample), the number of times a country became a democracy or reverted to autocracy, respectively, as well as (in robustness checks) the level of export flows in the year the exporter became a democracy (transformed using inverse hyperbolic sine) in the running line regression.

These multivariate running line regressions are local linear regressions of k nearest neighbours for the estimated treatment effect $\hat{\theta}_{ij}$ from the gravity equation for exporter i and destination j on a set of covariates. To mimic robust regression which attaches lower weight to outliers (Hamilton, 1992) we first compute weights from the regression of the pairwise treatment effect θ_{ij} on an intercept using an M-estimator (Rousseeuw & Leroy, 1987) before we run the multivariate running line regressions for all country-pairs with non-zero weight (typically over 93% of all pairwise estimates). All controls mentioned above are treated using local linear regression (and hence non-linearly), except the dummies for the start year of the pairwise regression sample, which are controlled linearly. Since we want to create a sense of the long-run relationship we group all estimates for countries with 30 or more years in democracy (20 in the sub-sample analysis for 1950-1989 and 1990-2014).¹⁶

¹⁶More specifically, we give them the length of treatment value of 32 (22) and add a further variable capturing *the deviation* from 32 years in democracy for all countries grouped in the 'long-run' and zero for all with fewer than 30

For inference, we adopt the principle of the nonparametric variance estimator of Pesaran (2006) and apply this to the running line regressions: $\widehat{\Omega}(\hat{\theta}^{MG,k}) = [N(N-1)]^{-1} \sum_{i=1}^k (\hat{\theta}_{ij}^k - \hat{\theta}^{MG,k})^2$, i.e. the variance for the average treatment effect of k nearest neighbours (similar number of years in democracy) is the mean of squared deviations of the individual treatment effect for pair ij from the local average treatment effect.

The result plots show the predicted treatment effect (y -axis) — predictions are based on all controls — for the number of years spent in democracy (x -axis). If treated country i has spent thirty years in democracy and is trading with twenty destinations, then there will be twenty markers (predictions) at the x -value of 30, and each marker is randomly perturbed to highlight this multitude. Filled (hollow) markers indicate that the confidence intervals for the local average treatment effect (local to the years of treatment) is statistically (in-)significantly different from zero at the 10% level.

4 Data

Data Sources: Gravity Model Variables We use bilateral trade data from TRADHIST (Fouquin and Hugot, 2016, version 4) for trade flows, exporter and importer GDP and Population, combined with CEPII data for FTAs and currency unions. Our temporal coverage is from 1950 to 2014.¹⁷ For the export flow data we code the ‘plausible zeros’ as zero rather than missing. Data on democracy is taken from the Varieties of Democracy (V-Dem) project: we primarily focus on their liberal democracy index but also explore the polyarchy and liberal component indices. These data are described in great detail in Boese (2019). In a robustness check, we adopt the definition of democracy by Acemoglu et al (2019), which combines information from Polity IV, Freedom House, and an in-depth study of the individual cases for regime change. We discard country pairs with fewer than 14 observations over time — this is by the necessity of the demands of our heterogeneous panel estimator.

Data Sources: Proxies for Deep Determinants For the heterogeneity analysis we adopt a range of measures as proxies for the ‘deep determinants’ of comparative development. For the disease environment aspect, we adopt two datasets related to malaria: the percentage share of the population at risk of malaria in 1965 from McCord et al (2017) and malaria ecology, a measure of the latent risk of malaria, from Kiszewski et al (2004). We also use historical data collated by Murray and Schaller (2010) on the prevalence of seven parasitic and infectious diseases, and data from Auer (2013) on early (18th century) disease environment, extended in Ertan et al (2016). For climate-related aspects of geography, we use data collated by Nunn and Puga (2012) on average latitude and land area in

(20) years of treatment.

¹⁷Earlier years are in principle available, but the heterogeneous coverage across countries creates much worse coverage for countries which remained autocracies throughout the sample period. This in turn affects the ability of the cross-section averages to capture unobserved common factors in these early years which then leads to the failure of the weak parallel trend type (Alpha) test we devise.

the tropical climate zone to construct dummies for low absolute latitude (the region between the Tropic of Cancer and the Tropic of Capricorn) and ‘some land in the tropics’. We further construct a dummy for no land in the temperate climate zone using information from Spolaore and Wacziarg (2013), from where we also collect information on whether countries are landlocked (unrelated to climate). From Andersen et al (2016) we take data on UV radiation intensity and information on frost days per year (originally constructed by Masters and McMillan, 2001).

Information on French legal origin is taken from La Porta et al (2008), from which we also construct a dummy for all countries which do not have British legal origin.

Proxies for culture relate to aspects of Hofstede’s concept of ‘individualism’ (good for innovation and growth) versus ‘collectivism’ (less good), and related proxies with broader country coverage from Gorodnichenko and Roland (2017): to blood type distance to the UK and US (the two most individualistic countries in the world). We further study the ability to communicate, proxied by language similarity, which we construct as country averages from dyadic data by Gurevich et al (2021). Our proxies capture the probability that two individuals picked at random from each pair of countries speak the same native language (cni) and a population-weighted proximity measure based on ‘linguistic trees’ which categorise languages (lp).¹⁸ Following the ideas of Max Weber, dummies for ‘no descendants from European settlers’ today and ‘no European settlers’ at the turn of the 20th century based on data collated by Spolaore and Wacziarg (2013) and Gorodnichenko and Roland (2017) are our further proxies for culture.

For aspects of colonial history we use the COLDAT database (Becker, 2019) from which we construct dummies for colonial experience, early colonisation (before the 19th century), and late independence (after the 1950s). The rationale of the latter two is to gauge ‘long exposure’ to colonial domination.

Data Transformations: Democracy We follow Boese-Schlösser and Eberhardt (2023) and adopt the full gravity data sample mean for liberal democracy (0.34) over 1950-2014 as the threshold for democratic regime change. We proceed in analogy for the polyarchy and liberal component measures. In robustness checks, we add either a quarter or half a standard deviation to this mean index value to adopt a much tighter definition of democratic regime change. The liberal democracy sample includes 761,637 observations for 15,771 country pairs (average T is 48). Of these 7,458 pairs are for regime changes in the exporter country, relating to 71 countries.

Data Transformations: Deep Determinants Our empirical setup builds on crude sample splits of deep determinants into ‘favourable’ and ‘unfavourable’ for economic development. For a range

¹⁸The DICL data by Gurevich et al (2021) are dyadic. However, to maintain the logic of our analysis by deep determinant (typically monadic), we are interested in and employ exporter i ’s *average* ability to communicate with all other countries, not their ability with specific importer j .

of proxies, this is straightforward by the nature of the data, e.g. French legal origin or no European settlers in 1900. For many of the proxies which are indices, we select the full exporter country sample median (i.e. the time-invariant values for 100 to 160 countries depending on the proxy) to split the sample in two. A notable exception here is 'Absolute Latitude' for which we adopt the Tropics of Capricorn and Cancer as cutoffs (which incidentally are close to the median absolute latitude).

5 Results

5.1 Main Results and Robustness Checks

Main Results Panel (a) of Figure 1 indicates the distribution of years in democracy for four different samples, defined by the destination: (i) all pairs, (ii) pairs where the importer is a democracy throughout the sample period, (iii) pairs where the importer experiences regime change during the sample period, and (iv) pairs where the importer is autocratic throughout. In this and all results plots by construction the country experiencing regime change is always the exporter i , and on the x -axis we can read off the total number of years the exporter country has spent in democracy. With exception of the democratiser sub-sample,¹⁹ around one-quarter of estimates are for exporters with ten or fewer years experience of democracy, 30-35% for those with 11 to 20 years, 24-28% for those with 21 to 30 years and 13-20% for those with more than 30 years of democratic experience after 1949. These figures suggest that the thousands of democracy estimates making up the running line predictions plotted are relatively uniformly distributed across these different periods in democracy, ruling out that differential sample size drives the results we report.

In panel (b) of the same figure we report the predictions for multivariate running line regressions for the four groups of importers over the period 1950-2014.²⁰ As is found in the existing literature on democracy and growth (Acemoglu et al, 2019), the initial period following democratic regime change is accompanied by economic 'hardship' or at least relative stagnation, in the sense that the causal effect of democratic regime change is negative and at times statistically significant: the average democratising exporter experiences around 20% decline in their exports relative to the control sample of never-democratisers (blue line and markers). Other than the 'tumultuous youth' experienced by many democracies (Gerring et al, 2005), at least some of this detrimental effect can be attributed to the crude way we (and the existing literature) define democratic regime change. It is notable how long this period of stagnation lasts: the average country has to wait 16 years until it experiences a positive significant treatment effect of regime change. Incidentally, the positive and significant treatment effect of regime change for *economic growth* in Boese-Schlosser and Eberhardt (2023), using the

¹⁹Here the youngest democracies only make up 7% of the sample, 24% of estimates are for countries with 11-20 years of experience, and 35% each for 21-30 and over 30 years, respectively.

²⁰The notes to Figure 1 report the Pseudo Alpha test results, which suggest that the weak parallel trend assumption crucial for identification is not rejected.

same liberal democracy definition, emerges after the identical number of years. Subsequently, returns to democracy rise steeply until around 22 years in democracy (around 40%), after which they regress somewhat. The median country in the full sample experiences 20 years in democracy, which according to our analysis leads to a 35% higher level of exports relative to the control sample of countries which stayed autocratic. The 'long-run' estimate for all exporters with 30 or more years in democracy is around 32%.

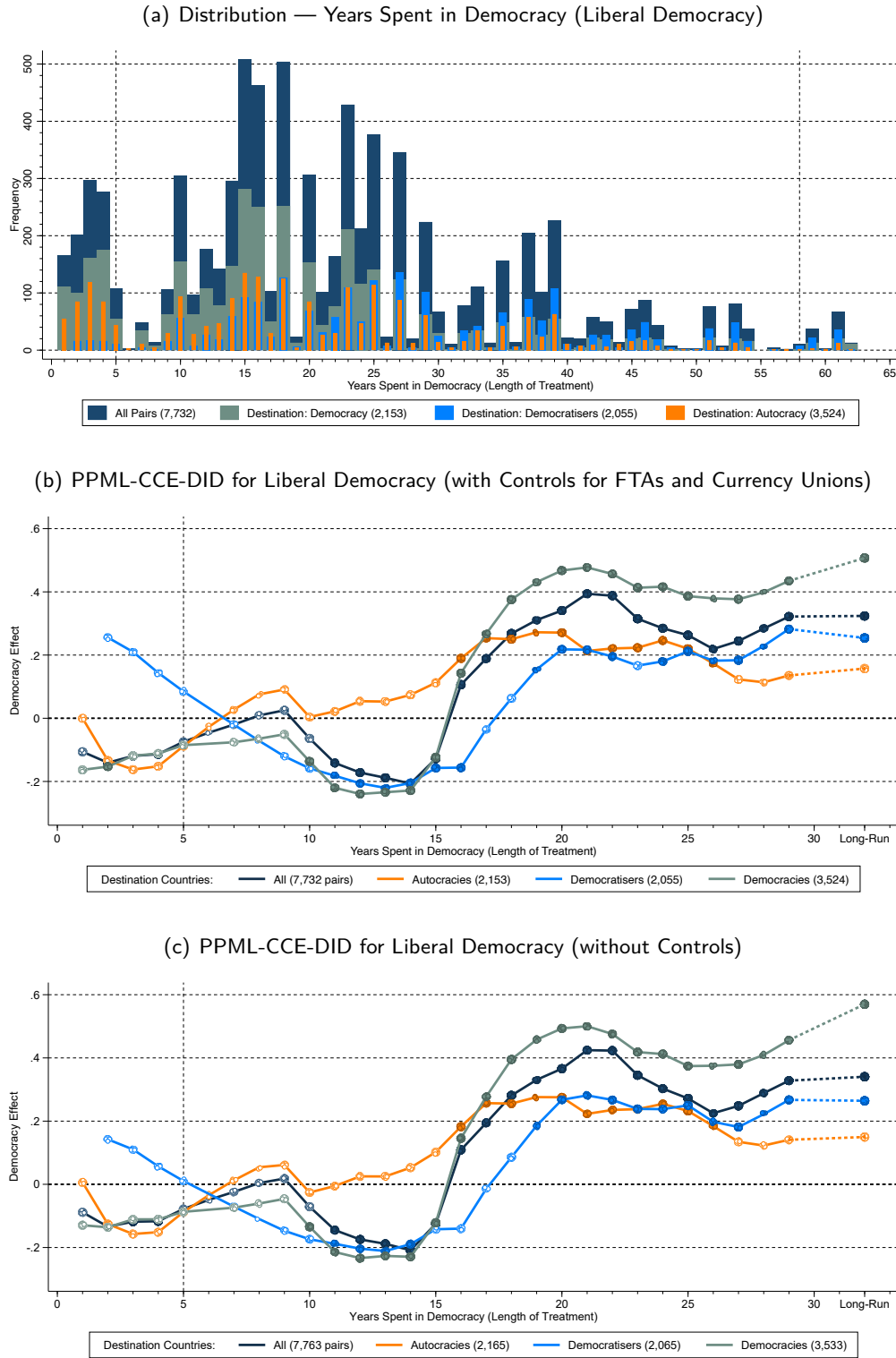
Shifting our attention to the sub-samples defined by importer regime type (always democracy, democratiser, always autocracy), we note the relatively moderate effects if the destination country is an autocracy throughout the sample period: at most 27% after 20 years in democracy, the long-run effect is clearly the lowest among the subsamples and statistically insignificant. Trade with democracies leads to a steeper increase after the decade and a half without positive effects, reaching almost 50% higher trade after twenty years. The long-run estimate is of similar magnitude. Finally, the results for trading with countries experiencing regime change at one point in the sample period leads to more moderate effects, initially very low and subsequently similar to those of autocratic trading partners, though the former's long-run effect is somewhat higher (22%) and statistically significant.

Our gravity equation includes measures for trade frictions at the dyadic level: dummies for free trade agreements and common currency arrangements. These are the primary objects of analysis under the current structural gravity model paradigm (what we called the 'straitjacket') and hence we naturally included them in our models. However, in the context of a treatment effects analysis, these could be seen as 'bad controls', in that democratic regime change may result in the increased propensity to enter free trade agreements (although the analysis by Baier and Bergstrand, 2004, in this regard found no evidence for 'institutional' characteristics affecting the likelihood of an FTA). Therefore, in panel (c) of Figure 1 we report the predicted running line plots by replicating our heterogeneous gravity regressions excluding the FTA and common currency indicators. As can be seen, the evolution paths are next to identical to those we reported above.

For these two specifications (with and without controls) we also provide robust mean treatment effect estimates by decade in democracy — see Appendix Figure B-1.²¹ These estimates suggest that regardless of the regime status of the trade partner, the first twenty years after a country experiences regime change the export effects are economically small and statistically insignificant. For 20-29 years estimates are around 25% (autocracies) and almost 40% (democracies, democratisers), while 30-39 years are associated with a stable 40% in the democratic trade partners and substantially lower and statistically insignificant effects if trade partners are autocratic. Effects for partners which are Democratisers lie inbetween.

²¹These estimates account for regime reversal but not for the year in which the country-sample starts.

Figure 1: Democratic Regime Change and Trade Flows — Effect by Years in Regime (1950-2014)



Notes: These plots present prediction paths from multivariate running line regressions, conditioning on years in democracy, number of democratisations and reversals to autocracy, respectively, and the data series start year. We group all democracy estimates for 30 or more years in democracy as the 'Long-Run' (in the multivariate running line regression we add an additional control, indicating the years in democracy minus 32 for those with 30 or more years, and zero for any country with fewer than 30 years). Medians length in democracy for all countries, (always) autocracies, democratisers and (always) democracies are 20, 16, 27, 20 years, respectively. The Pseudo Alpha tests for weak parallel trends between the control samples and the treatment sample for panel (b) have χ^2 statistics of 1.76 (p -value .18) and 1.81 (p -value .18) for exporter and importer control samples, respectively. A joint test results in a χ^2 statistics of 1.00 (p -value .31). For panel (c) these statistics are 1.00 (p -value .32) and 2.31 (p -value .13) for exporter and importer samples, 1.00 (p -value .32) for the joint test.

What is the significance of accounting for treated trade pairs (i.e. where an exporter experienced regime change) with zero trade flows throughout the sample period? As we had argued above, ignoring these would likely lead to a selection effect which would bias treatment estimates upwards. In Appendix Figure B-6 we report the running line plots based on PPML-CCE-DID estimates from models with and without the FTA and currency union dummies: these models essentially exclude the final line of equation (22) (bar the residual term ϵ_{ijt}). While the shape of the treatment effect-treatment length line is qualitatively identical, it is notable that (a) the different subsample estimates are much more similar to each other, suggesting that differential prevalence of zero trade flows across subsamples do indeed influence these results; and (b) that the long-run estimates at 80% (specification with controls) and 90% (specification without controls) are inflated compared with our estimates in Figure 1, indicating that our selection correction in the latter addressed the bias introduced when we ignore zero trade flow treated pairs.

Robustness We further estimated multivariate running line regressions (using the democracy effects from the heterogeneous gravity models) where we additionally control for the trade flow from exporter to destination in the year the exporter experienced democratic regime change (these values are transformed using the inverse hyperbolic sine function). Since our estimates are from heterogeneous parameter models, this can account for different growth benchmarks: for instance, the percentage increase in trade due to regime change from a low level of trade can be expected to be higher than that from an already higher level (convergence effect). Appendix Figure B-7 for the PPML-CCE regression with or without the FTA and Currency Union dummies shows patterns which are more or less identical to those in Figure 1.

All of the above results are based on taking a fairly arbitrary (mean) cut-off of the liberal democracy index to dichotomise democratic regimes. As a robustness check, we use the 1950-2014 mean of the V-Dem liberal democracy index plus one quarter or one half its standard deviation as the cut-offs for regime change. As Appendix Figure B-8 illustrates, we find much more common trajectories over the years spent in democracy, arriving at a 60% long-run effect. For the most conservative democracy threshold in panel (c), the statistically significant effects of democracy manifest themselves much earlier on, from 5 years in democracy. The sample where destinations are fellow democratisers now dominates the other subsamples in terms of the magnitude of the trade effect. This suggests that parts of the long (15 years) null effect in the main results can indeed be explained by the imprecise (*and ad hoc*) definition of the democracy dummy.

We also used the categorisation of Acemoglu et al (2019) which is based on Polity IV and Freedom House, in addition to detailed case analysis à la Papaioannou and Siourounis (2008). These data only cover 1960 to 2010, but despite 14 fewer sample years, the number of country-pairs with democratic regime change in the exporter country increases by 11% compared with our analysis

using the definition of regime change based on V-Dem. Appendix Figure B-9 shares characteristics with our main results (over a dozen years before treatment effects are positive significant, long-run effect at 60%, results by destination regime type are very similar), but also accentuates the mild concavity we observed in these with a very significant U-shaped trajectory between 17 and 29 years in treatment.

In summary, in the long run, the shift to democracy is associated with a sizeable increase in exports, on the order of 60-70% or around 2% per year spent in democracy. This effect manifests itself following a relatively lengthy period of lackluster performance for the first dozen or so years after regime change. Perhaps somewhat surprisingly, the trajectories of the treatment effect over years spent in a democracy are relatively similar whether importers are autocracies or democracies throughout the sample period. We now turn to the building blocks of liberal democracy.

5.2 Building Blocks of Liberal Democracy

Political scientists commonly adopt minimal definitions of democracy which build on Dahl's (1971) work and emphasise democracy as a "competitive struggle for the people's vote" (Schumpeter, 1942: 26). They argue that the accountability of a regime can only come from the power of the electorate to extend or withdraw the mandate of the leader(s): "Democracy is a system in which parties lose elections" (Przeworski, 1991: 10). Economists, in contrast, commonly put greater emphasis on the 'Northian' institutions (North, 1981; North and Weingast, 1989), a suite of civil rights (the 'rule of law') and constraints on the executive. These institutional building blocks ensure that the government is providing the right incentives and opportunities to private businesses ('getting the prices right'), which entails the reduction of market frictions (e.g. financial development) and the facilitation of transactions more generally, most notably foreign trade (Besley, 1995).

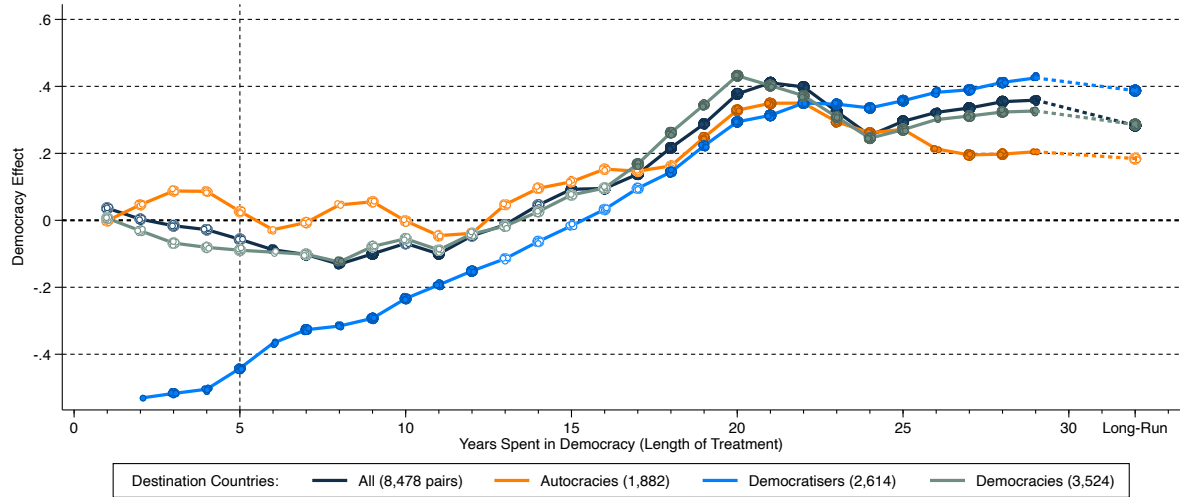
In our empirical analysis these different building blocks are represented by the V-Dem proxies for 'polyarchy' (electoral democracy) and the 'liberal component' — see Boese-Schlösser and Eberhardt (2023) for detailed discussion of these concepts and the operationalisation. As is the case for the 'liberal democracy' analysis above we dichotomise these two indices at their full sample mean to construct treatment dummies for 'regime' change.

For polyarchy, in panel (a) of Figure 2,²² we observe patterns similar to those in the analysis of liberal democracy above: slow emergence from stagnation (possibly a little faster here), and swiftly rising export effects after around 15 years which subsequently follow a U-shaped trajectory. The long-run effect at 40% higher exports is somewhat lower than for the liberal democracy analysis, but again very similar across the three subsamples.

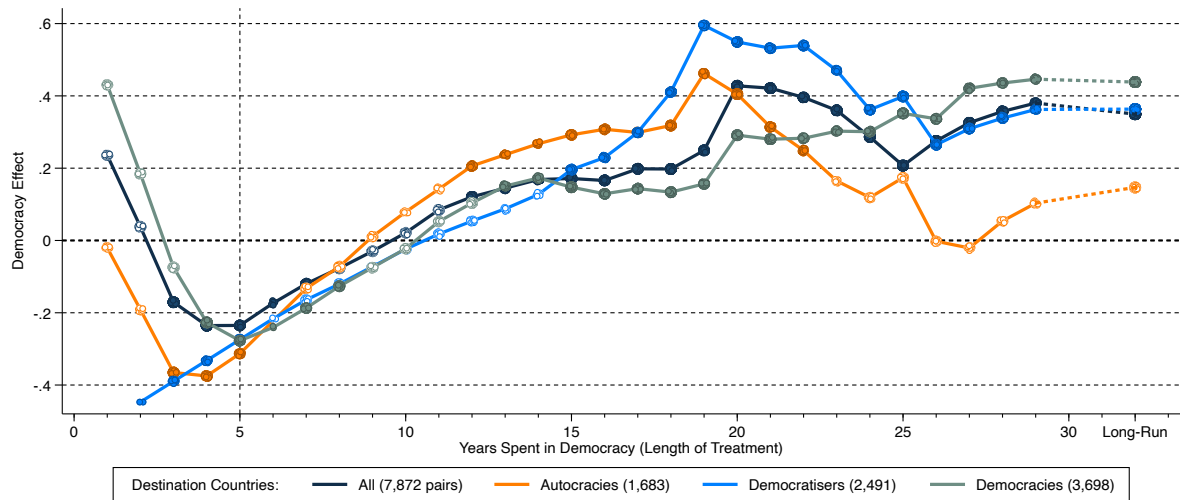
²²See Figure notes for details on the Pseudo Alpha specification test.

Figure 2: Beyond Liberal Democracy — Polyarchy and the Liberal Component

(a) Polyarchy (1950-2014)



(b) Liberal Component (1950-2014)



Notes: These plots present prediction paths from multivariate running line regressions, conditioning on years in democracy, number of democratisations and reversals to autocracy, respectively, and the data series start year. In panel (a) we study electoral democracy (polyarchy), in panel (b) the liberal component. The Pseudo Alpha tests for weak parallel trends between the control samples and the treatment sample for panel (a) have χ^2 statistics of 0.60 (p -value .44) and 0.33 (p -value .56) for exporter and importer control samples, respectively. A joint test results in a χ^2 statistics of 1.36 (p -value .24). For panel (b) these statistics are 0.28 (p -value .60) and 1.07 (p -value .30) for exporter and importer samples, 0.82 (p -value .37) for the joint test.

Our analysis in panel (b) of the bundle of institutions captured by the 'liberal component' suggests a peak of 60% higher trade after 20 years and overall comparable patterns to that in the analysis of polyarchy up until 30 years: here the long-run treatment effect is barely above 20%.

The most obvious conclusion from this analysis is that the sum of electoral-political (polyarchy) and politico-economic (liberal component) institutions is more than its constituent parts: factoring in electoral rights *and* the rule of law alongside executive constraints (liberal democracy) boosts the trade effect of regime change substantially more than either of these building blocks on their own. This aside, we see much less of the reversal of a positive trade effect among maturing liberal democracies (those with 20-30 years in a democracy) than if we limit the analysis to narrower institutions supporting elections or individual/property rights along with checks and balances.

5.3 Cold War and Post-Cold War Periods

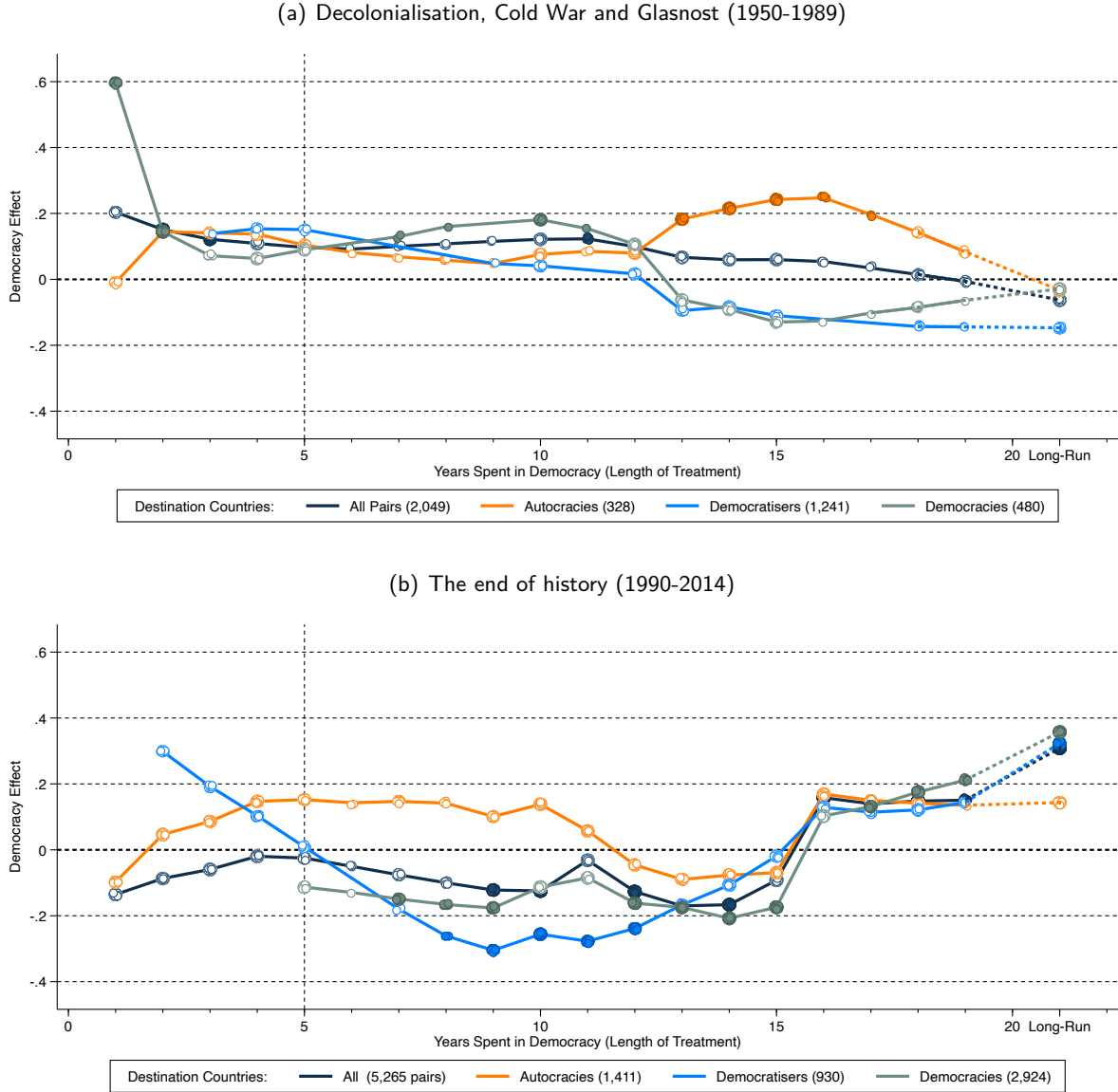
Recent work by Gokmen (2017) adopting the gravity model framework has demonstrated that the Cold War period of ideological strife before the collapse of the Soviet Bloc in the early 1990s saw cultural similarities and other country characteristics commonly acknowledged to have a substantial effect on trade flow volumes *suppressed* compared with the post-Cold War period. Ideology was a significant driver of trade, and the 'reduction' in ideology after 1989 meant the detrimental effects of cultural differences came to the fore during the later period, building effective trade barriers. Below, we investigate the trade effects of democratic regime change in two sub-periods demarcated by the end of the Cold War: 1950-1989 and 1990-2014. In our analysis we will adopt the identical empirical setup to the full sample analysis, with one small change: due to the shorter samples we designate twenty or more years spent in democracy as the 'long-run'.

In Panel (a) of Figure 3 we report the running line plots for the trade effect of democratic regime change limited to the 1950-1989 period.²³ Ignoring the effects for countries with very few 'treatment years' (fewer than 5 years), we can see no evidence for any tangible economic returns to democratic regime change. Long-run estimates are negative (except for destinations which are democracies) but statistically insignificant: substantially improving political and economic institutions did not result in any higher economic rewards in the form of increased exports.

In Panel (b) of the same Figure we study the period from 1990 to 2014. Again, we ignore the first few treatment years (since these are based on very few years in democracy relative to a large number of years in autocracy). The estimated effects for the full sample (navy line and markers) arrive at a positive and significant trade boost within a decade after regime change, but the overall trajectory is quite mooted and does not exceed 20%, even in the long run. Studying the different destination sub-samples, it is trade with autocracies (China?) which stands out, statistically significant and over 30%

²³As the notes to Figure 3 reports, the weak parallel trend assumption is not rejected in the Pseudo Alpha tests in both sub-periods.

Figure 3: Democratic Regime Change (Liberal Democracy) and Trade Flows — by Time Period



Notes: These plots present prediction paths from multivariate running line regressions, conditioning on years in democracy, number of democratisations and reversals to autocracy, respectively, and the data series start year. For Panel (a) see notes to Figure 1. For panels (b) and (c) we group all democracy estimates for 20 or more years in democracy as the 'Long-Run' (in the multivariate running line regression we add an additional control, indicating the years in democracy minus 22 for those with 20 or more years, and zero for any country with fewer than 20 years). The Pseudo Alpha tests for weak parallel trends between the control samples and the treatment sample for panel (a) have χ^2 statistics of 1.79 (p -value .18) and 1.09 (p -value .30) for exporter and importer control samples, respectively. A joint test results in a χ^2 statistics of 0.91 (p -value .34). For panel (b) these statistics are 3.56 (p -value .06) and 2.80 (p -value .09) for exporter and importer samples, 6.37 (p -value .01) for the joint test. The median number of years in democracy for the four groups are 9, 11, 0 and 11 for 1950-1989 and 15, 14, 16, and 14 for 1990-2014.

for most of the time after a decade in democracy. The effect of trading with democracies in contrast is insignificant throughout except for the 20% long-run effect (20 or more years in democracy). For democratisers, the sample size is substantially smaller and hence we should not read too much into the zero long-run estimate.

This analysis has confirmed that there were no significant trade effects of regime change in the period characterised by de-colonisation, the Cold War, and the collapse of the Soviet Union. Effects after 1990 are substantially smaller than for our full sample analysis, only around 20%, but we need to raise at least two important caveats: first, the period here is much shorter and includes the Global Financial Crisis in the latter part, thus naturally attenuating the potential for growth effects. Second, while the length of the pairwise time series is thus reduced, we still run the same demanding empirical specification (with 13 parameters to estimate) for this sub-period. In additional analysis (available on request) we confirm that a ‘plain vanilla’ version of the model without FTA and currency union dummies yields virtually identical results. The more modest regime change effect *on average* may be driven by greater heterogeneity in the regime change effect *across exporters*, which is what we investigate next.

5.4 Systemic Patterns in Cross-Country Treatment Effect Heterogeneity

Recent work by Eberhardt et al (2024) suggests that aggregate capital flow patterns²⁴ following democratic regime change are strongly correlated with geographic features related to disease environment and climate: they demonstrate that the boost to capital flows (total gross flows or FDI flows) from regime change is substantially attenuated for countries with ‘unfavourable’ geography (e.g. high malaria ecology, land in the tropical zone), whereas other ‘deep determinants’ related to legal origin, culture, and historical experience (colonialisation) do not produce similarly distinct patterns. Subsequent analysis by these authors suggests that while there appears to be a strong *correlation* between ‘geography’ and the effects of democratic regime change, it is likely that the explanation for these patterns can be sought in these countries’ history: countries in the ‘favourable’ geography sample with experience of colonisation have treatment effects much more in line with the treated countries in the ‘unfavourable’ geography sample.

Table 1 reports robust mean treatment effects based on PPML-CCE-DID gravity regressions where treatment and control samples are confined to identical characteristics in terms of a wide range of proxies for deep determinants related to geography, legal origin, culture, and history. The intuition is that we investigate the treatment effect of regime change for a country with specific structural characteristics (e.g. high historical disease environment or French legal origin) against a set of control countries with identical structural characteristics. This enables us to take the effect of

²⁴Their analysis is for aggregate gross total capital inflows or FDI inflows, rather than the dyadic trade flows we study in the present paper.

structural characteristics on regime change (e.g. the notion that certain types of ‘culture’ or ‘history’, perhaps related to having more European settlers due to a better historical disease environment, see Acemoglu et al, 2001) out of the equation and ask: how would democratic regime change in a country with good (or poor) deep determinants affect trade flows relative to a country with the same structural characteristics but which remained autocratic?

We focus on the average treatment effect on the treated (ATET) *per year spent in democracy*, estimated from the gravity model results using an M-estimator (Rousseeuw & Leroy, 1987) to weigh down outliers:

$$\begin{aligned} \hat{\beta}_{ij}^{Dem} = & \theta_0 + \theta_1 \text{Years in Democracy}_i + \theta_2 \text{Years in Democracy}_i \times \text{Deep}_i \\ & + \theta_3 \text{ih}(\text{Exports})_{ij}^{Dem} + \theta_4 \text{ih}(\text{Exports})_{ij}^{Dem} \times \text{Deep}_i + \Psi' \text{Start Year}_{ij} + \varepsilon_{ij}, \end{aligned} \quad (23)$$

where $\hat{\beta}_{ij}^{Dem}$ is the treatment effect of democratic regime change for exporter i from the PPML-CCE-DID regression of trade flows from i to j . Deep_i is a dummy for the deep (and immutable) determinant of comparative development and takes the value 1 if country i is in the ‘unfavourable deep determinant’ sample and zero otherwise. $\text{ih}(\text{Exports})_{ij}^{Dem}$ is the value of exports from i to j in the year country i experienced regime change — since this can be zero we adopt the inverse hyperbolic sine (ih) transformation for these values. This export term is *also* interacted with the ‘unfavourable deep determinant’ dummy. Finally, Start Year_{ij} is a set of dummies for the year the data series for trade from i to j starts.

These regressions are run separately by deep determinant (not indicated in the equation for ease of exposition) and we report the annual change in exports per year in democracy in the favourable ($\hat{\theta}_1$) and unfavourable ($\hat{\theta}_1 + \hat{\theta}_2$) deep determinant samples, respectively, alongside a p -value indicating whether these values are statistically significantly different from each other (based on the t -ratio of $\hat{\theta}_2$). Conditioning on the values of trade flows in the year of democratic regime change anchors these estimates for the annual democracy trade effects and allows us to compare the regime change effect across samples of favourable and unfavourable deep determinants.

In Table 1 (for the 1950–2014 time period) almost all estimates of the trade effect of democratic regime change per additional year in democracy in columns (1) or (2) are statistically significantly different from zero: we strike out an estimate if this is not the case, e.g. 0.1% for Early Disease Environment in column (1). Regardless of the geographic, legal, cultural, or historical characteristics, experiencing democratic regime change virtually *unambiguously* leads to higher exports compared with countries with the same characteristics but which remained autocratic. The magnitudes of the effects are typically sizeable, even for the lowest estimate below 1% per annum, since the average country experiences around twenty years in democracy: democracy does cause economic prosperity, including through the channel of increased exports.

Table 1: Deep Determinants and the Democracy-Trade Effect (1950-2014)

		(1)	(2)	(3)	(4)
		Unfav'ble	Deep	Determinant	
		0	1	Diff	p(Diff)
Geography					
Malaria Ecology	High	0.9%	1.4%	0.5%	0.07
Malaria Risk	High	1.4%	1.2%	-0.2%	0.55
Historical Disease Prevalence	High	1.8%	1.6%	-0.2%	0.48
Early Disease Environment (Auer)	High	-0.2%	0.6%	0.9%	0.00
UV Radiation	High	1.9%	0.2%	-1.6%	0.00
Absolute Latitude	Low	2.0%	0.7%	-1.3%	0.00
Landlocked	Yes	1.8%	1.1%	-0.7%	0.34
Few frost days per year	Yes	1.0%	-0.1%	-1.0%	0.00
Some land in Tropical Zone	Yes	1.4%	1.3%	-0.2%	0.58
No land in Temperate Zone	Yes	1.3%	1.0%	-0.3%	0.22
Unfavourable Geography	Yes	1.5%	0.9%	-0.6%	0.00
Legal Origin					
French Legal Origin	Yes	0.4%	1.4%	1.0%	0.00
Non-British Legal Origin	Yes	0.1%	1.9%	1.8%	0.00
Unfavourable Legal Origin	Yes	0.3%	1.7%	1.4%	0.00
Culture					
Collectivist	Yes	0.8%	1.7%	0.9%	0.00
Blood type distance to UK	High	0.7%	1.1%	0.5%	0.09
Blood type distance to US	High	0.9%	1.0%	0.2%	0.58
Avg Common Language Index	Low	0.7%	0.2%	-0.5%	0.07
Avg Language Similarity	Low	1.4%	0.3%	-1.1%	0.00
Zero Share European descendants	Yes	1.7%	0.9%	-0.7%	0.01
No European settlers in 1900	Yes	0.8%	1.1%	0.3%	0.31
Unfavourable Culture	Yes	1.0%	0.9%	-0.1%	0.16
History					
Colonial Experience	Yes	1.3%	0.9%	-0.4%	0.18
Early Colonisation (<c19)	Yes	0.9%	0.2%	-0.6%	0.07
Late Independence (>1959)	Yes	1.5%	1.1%	-0.4%	0.31
Unfavourable History	Yes	1.2%	0.9%	-0.3%	0.07

Notes: The table presents robust mean estimates of the export effect of democratic regime change per additional year in democracy, columns (1) and (2). These are derived from pooled models of the pairwise treatment effect regressed on (i) years in democracy, (ii) the start year of the pair sample, and (iii) the trade flow value (from exporter to importer) in the year the exporter experienced democratic regime change (transformed using the inverse hyperbolic sine). We distinguish robust mean estimates by proxies for deep determinants related to geography, legal origin, culture, and history. All of these proxies are dichotomous and coded so that a value of 1 indicates unfavourable deep determinants (high disease environment, French legal origin, etc), whereas 0 indicates favourable deep determinants. Column (2) reports the robust mean estimates for countries with unfavourable deep determinants, column (1) for favourable — all of these estimates are statistically significantly different from zero, except those highlighted using 'strikethrough'. Column (3) reports the difference in means (unfavourable minus favourable mean estimate), and column (4) whether this difference is statistically significant (p -value). We use darker shading for proxies which show statistical significance in the difference between unfavourable and favourable deep determinants in the democracy-trade effect.

Turning to the differences by deep determinant in column (3), we note that many specifications indicate statistically significant deviations in the export effect of democracy (p -values are provided in the final column). Countries with unfavourable geography by and large have lower democracy effects than their peers with favourable characteristics, frequently statistically significantly so. The estimates by legal origin suggest that countries with unfavourable French legal origin (argued to be restrictive to the conduct of business and financial development in general) actually experience substantially higher trade growth following regime change. Estimates by culture and colonial experience once again suggest, by and large, that ‘unfavourable’ characteristics translate into a worse economic dividend of democratic regime change, much more substantially so in the case of colonial experience.

There are interesting differences and parallels with the capital flow analysis in Eberhardt et al (2024): while the strong geographic patterns in their average treatment effects also point to large attenuation for countries with ‘unfavourable’ geography, they do not find sizeable differences in the analysis of legal, cultural or historical characteristics.

However, as our full sample results above indicated, this investigation ignores the serious distortion of trade before the end of the Cold War with the collapse of the Soviet Union in the early 1990s, when ideology was a strong determinant of trade flows. We run separate analyses for the post-Cold War sample in an identical fashion and present our results in Table 2. The vast majority of estimates in columns (1) and (2) are still positive and significant (10 out of 46 mean estimates are negative and/or insignificant, compared with 8 for the 1950-2014 analysis), but the patterns and magnitude of differences have shifted somewhat: (a) we mostly see lower trade effects of countries with unfavourable geography, which on average are larger (-1.3%) than for the full sample (-0.6%); (b) unfavourable legal origin is now associated with worse trade effects and (c) cultural characteristics lead to on average larger deviations for those countries less culturally aligned (-0.7% compared with -0.1% in the full sample). Finally, (d) although quite mixed in terms of statistical significance, the gap between countries with colonial experience and others has switched sign (on average 0.5% rather than -1.1% in the full sample) and suggests higher trade effects of regime change for these structurally disadvantaged countries.

We can link these findings to Gokmen’s (2017) which was exclusively about trade flows, not trade flows following a significant improvement in institutional quality, in that the more recent post-Cold War period saw a stronger effect of division by culture, geography, or legal origin: countries with characteristics judged in the existing literature as unfavourable for economic development seem to not only fare worse relative to their ‘favoured’ peers, but this gap is greater in magnitude if we exclude the ideology-driven period of the Cold War from the analysis. Overall, we can conclude that structurally disadvantaged exporters have substantially lower trade effects from democratic regime change, with culturally similar, geographically favourable and legally common-law nations reaping

Table 2: Deep Determinants and the Democracy-Trade Effect (1990-2014)

		(1)	(2)	(3)	(4)
		Deep Determinants			
		0	1	Diff	p(Diff)
Geography					
Malaria Ecology	High	0.8%	1.4%	0.5%	0.49
Malaria Risk	High	2.4%	2.9%	0.5%	0.49
Historical Disease Prevalence	High	2.6%	1.4%	-1.2%	0.14
Early Disease Environment (Auer)	High	3.5%	2.4%	-1.1%	0.12
UV Radiation	High	2.6%	1.0%	-1.6%	0.04
Absolute Latitude	Low	2.3%	0.0%	-2.3%	0.00
Landlocked	Yes	2.1%	0.8%	-1.3%	0.19
Few frost days per year	Yes	3.2%	-0.2%	-3.4%	0.00
Some land in Tropical Zone	Yes	1.3%	3.3%	1.9%	0.01
No land in Temperate Zone	Yes	2.2%	1.5%	-0.7%	0.30
Unfavourable Geography		2.3%	1.4%	-0.6%	0.00
Legal Origin					
French Legal Origin	Yes	-0.3%	1.9%	2.2%	0.00
Non-British Legal Origin	Yes	0.8%	2.0%	1.1%	0.12
Unfavourable Legal Origin		0.2%	2.0%	1.8%	0.00
Culture					
Collectivist	Yes	5.7%	-1.0%	-6.7%	0.00
Blood type distance to UK	High	1.5%	1.2%	-0.3%	0.66
Blood type distance to US	High	1.8%	1.6%	-0.1%	0.84
Avg Common Language Index	Low	3.4%	0.8%	-2.6%	0.00
Avg Language Similarity	Low	1.9%	0.3%	-1.6%	0.02
Zero Euro descendants	Yes	3.9%	0.5%	-3.4%	0.00
No European settlers in 1900	Yes	1.1%	1.3%	0.2%	0.80
Unfavourable Culture		2.4%	0.8%	-1.6%	0.00
History					
Colonial Experience	Yes	0.7%	2.0%	1.3%	0.15
Early Colonisation (<c19)	Yes	1.3%	2.5%	1.2%	0.33
Late Independence (>1959)	Yes	3.3%	0.9%	-2.4%	0.00
Unfavourable History	Yes	2.1%	1.7%	-0.5%	0.29

Notes: The table presents robust mean estimates of the export effect of democratic regime change per additional year in democracy — we limit the sample to the post-Cold War period (1990-2014). All mean estimates are statistically significantly different from zero at the 10% level, except those which are struck out. For details see notes to Table 1. Medians (means) for the underlying pairwise regressions are xx (xx.x) years in democracy for all samples (good or poor deep determinant for any deep determinant), 25th and 75th percentiles are xx and xx years in democracy, respectively, for all samples. There are on average 2,200 (2,900) pairs in the samples with a deep determinant value of 0 (1).

higher democratic dividends. This deviation is more marked when we limit the sample to countries shifting to democracy after the end of the Cold War, in line with Gokmen's (2017: 449) findings of "cultural determinants... as a major impediment to international trade". The relative trade boost for countries which experienced colonialism admittedly is empirically fragile, but provides a small counterweight to these pessimistic conclusions.

6 Concluding Remarks

In this paper we have introduced an empirical approach which seeks to both satisfy the rigour of the structural gravity model and enable the analysis of monadic (country-specific) variables within a 'heterogeneous' gravity model. This opens up the opportunity to study a vast array of trade determinants which the current literature had to ignore in its adherence to the empirical straitjacket of the pooled gravity equation. We employ this new machinery to investigate one of the most pertinent questions in recent decades, namely that of the impact of institutional change on economic prosperity: does democracy cause greater trade flows?

Our first set of results for the full sample from 1950 to 2014 provide an answer in form of a resounding 'yes' to this question: two decades after democratic regime change the average country can expect high (and subsequently stable) economic benefits in form of a 60% increase in exports. This effect is surprisingly similar whether destination countries are democracies, autocracies or themselves experience regime change during the sample period.

However, this finding masks substantial heterogeneities over time and across countries. If we limit the sample to 1950-1989, the period of the Cold War when ideological and geopolitical factors were the dominant determinants of trade flows, we can see that regime change had virtually no economic effect. Causal effects of regime change after the Cold War are markedly lower than over the full sample period, at best 20% higher than in the counterfactual sample.

Economic benefits from regime change also follow distinct patterns of geographic, cultural, and historical bias. In line with existing research, we find that these non-economic factors have come to the fore during the post-Cold War period. While regime change is unambiguously associated with higher export flows, countries which have unfavourable geography, comparatively distant culture, or French legal origin have substantially lower effects than their better-placed peers. Results for the historical experience of colonialism are quite mixed, but it would appear that the relatively worse trade effects in the full sample were primarily driven by the Cold War period, while after 1989 these countries may to some extent have been able to 'catch up'.

This is just a first attempt at highlighting the heterogeneous causal effects of institutional changes in the context of the empirical gravity model. Given the rich literature in political economy and political science, we expect the PPML-CCE-DID methodology we introduce to offer important

insights in this regard in future research. The wider implications of a heterogeneous gravity model, allowing for the study of any time-varying monadic variable, has of course significant potential to inform policy on a range of trade determinants which previously had to be ignored for the sake of rigorous empirical identification.

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